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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-26

Experiment 26. Best First Search Algorithm

Aim

To implement the **Best First Search algorithm using Prolog**.

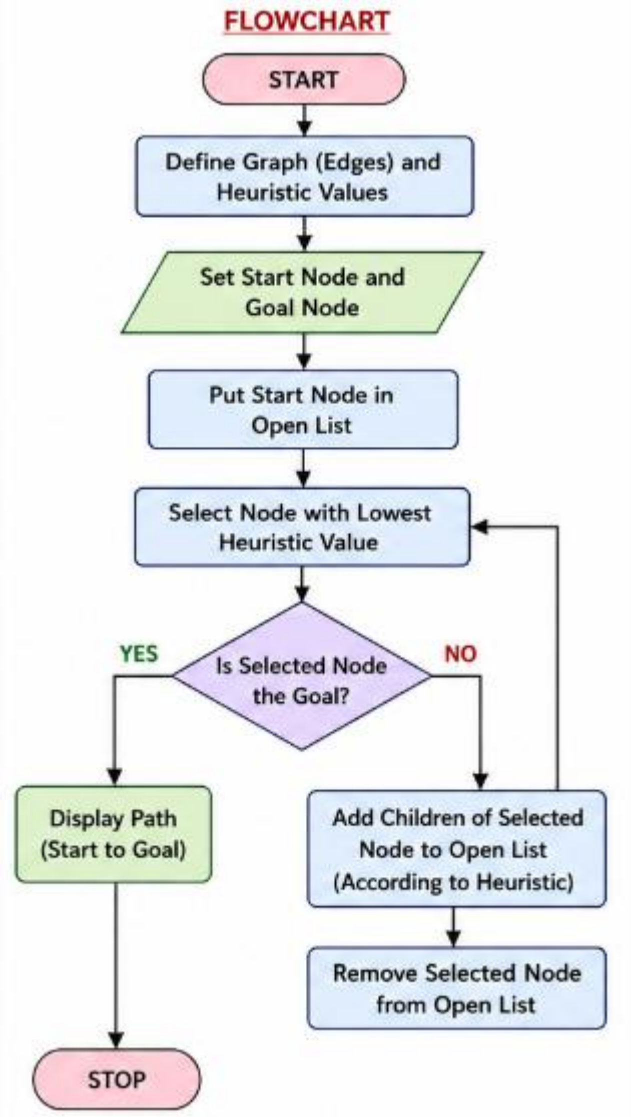
Objectives

- To understand heuristic-based search.
- To find a path from a start node to a goal node.
- To select the node with the lowest heuristic value.

Algorithm

1. Define the graph and heuristic values.
2. Set the start and goal nodes.
3. Select the node with the lowest heuristic value.
4. Expand its neighbouring nodes.
5. Repeat until the goal is reached.
6. Display the path.

Flowchart



Code

edge(a, b).

edge(a, c).

edge(b, d).

```
edge(c, e).

edge(d, goal).

edge(e, goal).


h(a, 4).

h(b, 2).

h(c, 3).

h(d, 1).

h(e, 2).

h(goal, 0).


best_first(Start, Goal, Path)
:-
    search([Start], Goal,
Path).
```

```
search([Goal|_], Goal,
[Goal]).


search([Current|Rest],
Goal, [Current|Path]) :-
   findall(H-Next,
         (edge(Current,
Next), h(Next, H)),
         Children),
   sort(Children, Sorted),
   get_nodes(Sorted,
Nodes),
   append(Nodes, Rest,
NewList),
   search(NewList, Goal,
Path).


get_nodes([], []).
```

```
get_nodes([_-Node|T],
[Node|R]) :-

    get_nodes(T, R).



main :-

    Start = a,

    Goal = goal,

    best_first(Start, Goal,
Path),

    write('Start: '),
write(Start), nl,

    write('Goal: '),
write(Goal), nl,

    write('Path: '),
write(Path), nl.



:- initialization(main).
```

Output

Result

Start: a

Goal: goal

Path: [a,b,d,goal]

Conclusion

Thus, the **Best First Search algorithm was successfully implemented using Prolog**.